



2023

COMPUTER SCIENCE — MDC

Paper : CC-1

Full Marks : 75

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer **question no. 1** and **any three** questions from **Section - A** and **any five** questions from **Section - B**.

1. Answer **any five** of the following : 2×5
- (a) Write two differences between low-level and high-level languages.
 - (b) Find out the octal representation of $(1F5 \cdot A2)_{16}$.
 - (c) Prove that $X + Y \cdot Z = (X + Y) \cdot (X + Z)$, where X, Y and Z are Boolean variables.
 - (d) Design a two input AND gate using a 2×1 multiplexer.
 - (e) What is a common anode type seven segment display?
 - (f) State the advantages of using synchronous counters over asynchronous counters.
 - (g) Mention the differences between sequential and combinational circuits.
 - (h) What is Delay flip-flop?

Section - A

2. Explain the working of a 4-bit binary adder-subtractor circuit with a neat diagram. 5
3. What is a magnitude comparator? Design a 1-bit comparator using NAND gates. 2+3
4. Convert a D flip-flop into a JK flip-flop. 5
5. (a) What do you mean by manchester code?
(b) What are the differences between weighted and non-weighted code?
(c) Convert $(9CB)_{16} = ()_2 ? = ()_8$? 1+2+2
6. (a) Design a D flip-flop.
(b) Convert it to Toggle flip-flop. 3+2

Please Turn Over



Z(1st Sm.) Computer Science-MDC/CC-I/CCF

(2)



Section - B

7. (a) Define half adder.
(b) Implement a full adder using two half adders.
(c) Why do we call decoders as min-term generators? Explain with an example. 2+3+5
8. (a) State De-Morgan's theorems.
(b) Convert into canonical POS. $Y = AB + BC$.
(c) Simplify the expression given below and draw the diagram by NAND gates :
 $Y = \Sigma_m(1, 2, 6, 7, 8, 13, 14, 15) + \Sigma_d(0, 3, 5, 12)$. 2+3+5
9. (a) Define full subtractor.
(b) Implement a full subtractor using a 4×1 multiplexer.
(c) Why demultiplexers are called data distributors?
(d) Design an even parity checker. 2+3+3+2
10. (a) Subtract 24_{10} from 14_{10} by 2's Complement method.
(b) Fill in the blank : $(17)_{10} + (21)_8 - (0F)_{16} = (?)_2$.
(c) State the differences among Super, Mainframe, Mini and Personal Computer. 3+3+4
11. (a) Define propagation delay and noise margin in a logic gate.
(b) Convert $(10110111)_2$ into gray code.
(c) Design a Johnson Counter using D flip-flops. 4+2+4
12. Design 32 to 1 multiplexer using 8 to 1 multiplexers. — Explain with truth table. 10
13. (a) Design a priority encoder using basic logic gates and give the truth table.
(b) Realize a 2 to 4 decoder using NAND logic gates. 5+5
14. (a) What is the truth table of a JK latch? From that deduce the characteristic equation, hence draw a JK latch using NAND gates.
(b) Define the race around condition of JK flip-flop and hence explain how master-slave JK flip-flop overcomes this issue. 5+5
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